Fiscal Year:	FY 2021	**Task Last Updated:**	FY 03/16/2021
PI Name:	Roberts, Donna M.D.		
Project Title:	Human Cerebral Vascular Autoregulation and Venous Outflow in Response to Microgravity-Induced Cephalad Fluid Redistribution		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	29425-8905	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
Start Date:	05/16/2013	**End Date:**	09/30/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2020 per NSSC information (Ed., 07/18/22) NOTE: End date changed to 10/1/2020 to accommodate final reporting and bibliographic information (Ed., 3/17/21) NOTE: End date changed to 9/30/2020 per PI (Ed., 9/30/19) NOTE: End date changed to 9/30/2019 per NSSC information (Ed., 2/26/19)		
Key Personnel Changes/Previous PI:	February 2018 report: added Dr. Bloomberg as Co-Investigator.		
COI Name (Institution):	Brown, Truman Ph.D. (Medical University of South Carolina) George, Mark S M.D. (Medical University of South Carolina) Bloomberg, Jacob Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX13AJ92G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE this is continuation of "Human Cerebral Vascular Autoregulation and Venous Outflow in Response to Microgravity-Induced Cephalad Fluid Redistribution" grant #NNX13AC32G, which ended 5/15/2013; Donna Roberts, M.D., continues as Principal Investigator. According to the NASA Human Research Roadmap (Risk of Microgravity-Induced Visual Impairment/Intracranial Pressure; Gaps VIIP2, VIIP4, VIIP5, VIIP8) and further expounded upon in the Visual Impairment Intracranial Pressure (ICP) Summit Report (May 2011), an essential gap exists in our knowledge of intracranial pressure regulation in spaceflight [Ed. note September 2019: Risk changed in November 2017 to Spaceflight Associated Neuro-ocular Syndrome (SANS); see https://]. This is exemplified by the fact that there are few published studies investigating the structural and functional changes occurring as a result of long-term spaceflight and there is almost no data concerning cerebral autoregulation in astronauts following return from space. Only two published studies exist investigating cerebral autoregulation in astronauts using an indirect method for measuring cerebral blood flow (CBF) at the level of the major arteries of the circle of Willis. Thus, although cerebrovascular hemodynamics plays a crucial role in ICP regulation and also in the response to orthostatic stress, we have almost no knowledge of the state of cerebral autoregulatory mechanisms during spaceflight. The operational relevance of this gap according to the NASA Human Research Roadmap is that elevated ICP and likely associated vision disturbances due to microgravity exposure are serious health risks for astronauts. Furthermore, this is an even greater concern for potential longer duration missions such as to the moon or Mars where decreased vision may impact mission performance and may even require early mission termination. Permanent visual losses may result in lifetime disability issues. Almost certainly, during long-term flights, as the hydrostatic pressure gradients due to gravity are abolished with resultant large cephalad fluid shifts, an intracranial adaptive response occurs. Thus direct measurements of cerebral fluid shifts and cerebral hemodynamics as proposed in this study will allow us to more fully characterize the intracranial compartment guiding the development of preventive measures, such as specific pharmacological countermeasures, to combat the development and sequelae of increased ICP during future missions. In order to fully characterize intracranial adaptation to long-term microgravity, we will use magnetic resonance imaging (MRI) which is the terrestrial gold standard. MRI allows direct visualization and functional characterization of the intracranial contents. Indeed, MR imaging is the clinical standard of care in evaluating patients with idiopathic intracranial hypertension. We will use a suite of advanced MRI sequences to: (1.) characterize weightlessness-induced intracranial compartmental fluid volume changes at the macroscopic (brain, cerebrospinal fluid-CSF) and vascular (arterial and venous) levels following adaptation to long-term spaceflight and (2.) assess for potential resultant hemodynamic changes (cerebral perfusion) in the brain. This proposal represents the most relevant and advanced suite of MRI techniques for the characterization of intracranial compartmental fluid and hemodynamic changes induced by long-term microgravity. The specific sequences will be carried out by the recognized experts in each area and in most cases, the developer of the specific technique. It is imperative to obtain this data in order to have an understanding of what is happening to the brain and orbits while living and working in microgravity. We are only now realizing the consequences of this lack of knowledge. This data will serve as the basis for the development of fundamental theories describing intracranial adaptation to microgravity. It may help establish a risk stratification strategy based on intracranial anatomy, dural venous sinus and jugular vessel outflow, and cerebral vascular autoregulation. Importantly, it will serve as a basis for interpretation of in-flight data collected via other techniques such as ultrasound, the primary in-flight imaging modality.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Idiopathic intracranial hypertension, or pseudotumor cerebri, is a poorly understood clinical condition of unknown etiology. It has a reported incidence of between 1 and 5 cases per 100,000 people in the general population, although the true incidence may be larger. There is a strong female predominance. Patients typically present with disabling headaches and vision changes. In more extreme cases, injury to the optic nerve can occur and patients can experience vision loss. Upon testing, these patients have increased intracranial pressure and swelling of the optic nerves. Similar symptoms can occur in astronauts following long-term spaceflight. This study will provide a unique opportunity to investigate the underlying pathophysiology of abnormal intracranial pressure regulation which will be applicable to those who suffer from this Earth-based condition. This research may also shed light on the pathophysiology of other Earth-based disease processes associated with abnormal cerebrospinal fluid (CSF) flow or venous outflow obstruction such as obstructive hydrocephalus, normal pressure hydrocephalus and multiple sclerosis, and pathologies associated with altered cerebral autoregulation such as traumatic brain injury.
Task Progress:	During this reporting period, we collaborated with another investigator (Dr. Seidler) in a jointly sponsored NASA-German space agency bedrest study performed at the Envihab facility in Cologne, Germany known as the VaPER study. We assessed cerebral perfusion in the VaPER participants using arterial spin labeling (ASL) MRI before, during, and after long-term head down tilt bedrest in a CO2 environment (head down tilt (HDT)+CO2)) as an analog for spaceflight. Our results were quite surprising. As hypercapnia is a potent cerebrovasodilator, we had hypothesized that HDT+CO2 would result in an increase in cerebral blood flow. Instead we found the opposite – by bed rest day 7 there was a 25.1% global decrease in cerebral blood flow compared with baseline. Additionally, there was a significant time-by-SANS interaction indicating that the change in perfusion over time varied as a function SANS status. The non-SANS group significantly decreased perfusion between BDC-13 and HDT7 ($p = .02$, 95% CI: 3.07 - 36.08, $d = 1.92$, very large effect) and between measurements at baseline and HDT29 (BDC-13: $p = .01$, 95% CI: 3.64 – 31.99, $d = 1.86$, very large effect; BDC-7: $p = .04$, 95% CI: 0.84 – 31.63, $d = 1.50$, very large effect). The SANS group significantly decreased perfusion between baseline and HDT7 (BDC-13: $p = .003$, 95% CI: 9.08 – 45.24, $d = 2.42$, huge effect; BDC-7: $p = .009$, 95% CI: 7.54 – 56.71, $d = 2.58$, huge effect). The SANS group showed the lowest perfusion at HDT7. However, they then experienced a significant increase in perfusion at HDT29 compared with HDT7 ($p = .03$, 95% CI: 3.27 – 39.92, $d = 2.07$, huge effect), R+5 ($p = .003$, 95% CI: 10.19 – 48.76, $d = 2.43$, huge effect), and R+12 ($p = .005$, 95% CI: 9.95 – 55.12, $d = 2.34$, huge effect). At HDT7, there was a strong trend for the SANS group to have decreased perfusion values compared with the non-SANS group ($p = .05$, 95% CI: -0.19 to 16.11, $d = .94$, large effect). At HDT29, the SANS group had significantly greater perfusion than the non-SANS group ($p = .04$, 95% CI: 0.33 to 13.07, $d = 1.01$, large effect).

	These results represent the first MRI assessment of cerebral perfusion in a spaceflight analog. In this study, we show that the combination of strict HDT along with chronic 0.5% CO2 results in altered cerebral perfusion that is SANS dependent. These results underscore the need for further studies to determine if the changes in perfusion documented here are clinically relevant, whether or not similar changes occur during spaceflight, and whether or not altered perfusion contributes to the development of SANS or headaches among astronauts. Furthermore, along with our NASA and international colleagues, we have been advocates for establishing collaborations among investigators both at NASA and internationally (Roberts, et al., Lancet Neurology, 2020--See Bibliography section). We believe these collaborative efforts will be crucial in elucidating the underlying mechanisms and clinical features of the intracranial component of SANS. Our project team participated in the annual NASA Human Research Program Investigators Workshop held virtually in February 2021. As an outreach effort, Dr. Roberts participated in the South Carolina Space Grant Consortium Summer Research Program, by mentoring summer students (3 undergraduate, 2 medical students, and 2 residents) who assisted with analysis of data for this study.
Bibliography Type:	Description: (Last Updated: 01/15/2026)
Articles in Peer-reviewed Journals	Roberts DR, Stahn AC, Seidler RD, Wuyts FL. "Towards understanding the effects of spaceflight on the brain." Lancet Neurol. 2020 Oct;19(10):808. Correspondence. https://doi.org/10.1016/S1474-4422(20)30304-5 ; PMID: 32949538 , Oct-2020
Articles in Peer-reviewed Journals	Roberts DR, Collins HR, Lee JK, Taylor JA, Turner M, Zaharchuk G, Wintermark M, Antonucci MU, Mulder ER, Gerlach DA, Asemani D, McGregor HR, Seidler RD. "Altered cerebral perfusion in response to chronic mild hypercapnia and head-down tilt bed rest as an analog for spaceflight." Neuroradiology. 2021 Aug;63(8):1271-81. https://doi.org/10.1007/s00234-021-02660-8 ; PubMed PMID: 33587162 , Aug-2021
Articles in Peer-reviewed Journals	Roberts DR, Inglesby D, Brown TR, Collins HR, Eckert MA, Asemani D. "Longitudinal change in ventricular volume is accelerated in astronauts undergoing long-duration spaceflight." Aging Brain. 2021 Jun;1:100017. https://doi.org/10.1016/j.nbas.2021.100017 , Jun-2021
Articles in Peer-reviewed Journals	Tidwell JB, Taylor JA, Collins HR, Chamberlin JH, Barisano G, Sepehrband F, Turner MD, Gauthier G, Mulder ER, Gerlach DA, Roberts DR. "Longitudinal changes in cerebral perfusion, perivascular space volume, and ventricular volume in a healthy cohort undergoing a spaceflight analog." AJNR Am J Neuroradiol. 2023 Sep;44(9):1026-31. https://doi.org/10.3174/ajnr.A7949 ; PMID: 37562828; PMCID: PMC10494950 , Sep-2023